DOCUMENT TITLE	REVISION
SRP-043 – Front Suspension – Conceptual Design Report	0

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Ryan Zi Seng Chuah z5335020	15/06/2024

Department	Project
Vehicle Dynamics	Front Suspension

REVISION HISTORY

Rev.	Change Log	Author/s	Date
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1. Projects

1.1 Project Scope

SRP – 041 will develop and implement the suspension subsystem acting within the larger Vehicle Dynamics subsystem for the SR8 project. The suspension subsystems involve the development of the suspension geometry, as well as the spring and damper system on for SR8. This development includes optimizing the subsystem for performance and the vision of the SR8 project.

Terms, Definitions, and Abbreviations

Table 1: Terms, Definitions, and Abbreviations

Abbreviation/Terms	**Definition**
TPM	Technical performance measures
SR8	Sunswift 8

1.2 Responsible Departments

Vehicle Dynamics (Responsible)

1.3 Secondary/Supporting Departments

Chassis & Bodywork (Supporting)

1.4 Project Interfaces

SRP – 048 will interface with the following departments:

Table 2: Terms, Definitions, and Abbreviations

Relevant Departments	**Description**
SRP – 001 Crash Structures	Ensure that crash structures do not interfere with suspension movement throughout full range of motion.
SRP – 002 Exterior Design	Ensure that crash structures do not interfere with suspension movement throughout full range of motion.
SRP – 004 Occupant Cell	Ensure that crash structures do not interfere with suspension movement throughout full range of motion.
SRP – 006 Subframes	Suspension is mounted to the subframes. Mounting points must be clearly defined
SRP – 010 Test Mule	Test mule has accurate dimensions (pickup points and dimensions) for effective testing and optimization.
SRP – 011 Active Aero	If active aero is included, the suspension stiffness must be able to account for the increased compression loads.

SRP – 035 Performance optimisation	Suspension performance can be adjusted to suit conditions, optimize damping ratios, and minimize energy generation (if included).
SRP – 038 Brakes	Ensuring pick up points defined by suspension allow adequate space for brake packaging and uprights
SRP – 040 Steering	Ensure that the suspension geometry can accommodate the required steering inputs and provides the desired tire contact patch
SRP – 042 Wheels	Ensure that there is no clearance issues with the wheel and tire throughout the entire suspension travel

1.5 Assigned Team Members

- Ryan Zi Seng Chuah (Project Lead)

- James Crisp

2. Design Factors

2.1 Lifecycle Analysis

The suspension system of the car should last the lifetime of the vehicle, with certain serviceable components being replaced at required intervals. Components expected to last the lifetime of the vehicle include the wishbones, push rods, chassis mounts and springs. The replaceable components include bushing and ball-joints, while serviceable components include dampers.

Rubber bushings have a life expectancy of approx. 80,000km, while poly-urethane bushings can last the lifetime of the car assuming they are lubricated once every 5 years.

Ball joints can go roughly 100,000 km before they require replacement while spherical bearings require constant lubrication and have much shorter total lifespans.

2.2 Risk Analysis

Major concerns related to suspension are related to the failure of critical components such as control arms, pushrods, dampers, springs etc. There is also the gradual failure of joints due to wear such as rubber bushings or ball joints. This gradual wear can result in a car that does not respond correctly to driver inputs due to excessive play in joints. A full list of identified risks is found in Table 2 below.

Table 3: Risk Analysis Table

Risk	Response Category	Mitigation Strategy
Suspension Hardware		
Ball joint wear	Accept	Regular servicing/inspections which are included as a vehicle inspection check. See lifecycle for lifespan of these components.
Bushing Wear	Accept	Regular servicing/inspections which are included as a vehicle inspection check. See lifecycle for lifespan of these components.
Damper leak	Mitigate	Replace/Service the faulty equipment when leak identified
Critical Failure Modes		
Composite spring failure	Mitigate	Regular inspections which are included as vehicle inspection check. Replace spring if any damage to material
Control arm failure	Mitigate	Regular inspections which are included as vehicle inspection check. Replace control arm if any damage to material
Pushrod failure	Mitigate	Regular inspections which are included as vehicle inspection check. Replace push rod if any damage to material
Loose hardware	Mitigate	Regular inspections which are included as vehicle inspection check. Correct use of fasteners and thread locking compounds.
Project Unknowns		
Composite spring unavailable in time	Avoid	The car will be able to use both conventional coil overs or a transverse leaf spring to account for the experimental nature of the leaf spring development.

Risk	Impact	Response

2.3 Innovation Analysis

The actual suspension geometry does not have any aspects that allow for innovation, it has been well studied over the years and there is no need to reinvent the wheel. The extent of innovation in the suspension geometry has been limited to material choices, the suspension geometry of Double-wishbone is present on many high performance cars and as such is low in innovation, but also very low in risk. The decision on materials for the suspension geometry will not be made until the forces acting through the structural members are known. Instead, the innovation of the suspension project is focused on the spring and damper system.

The composite transverse leaf spring is an interesting concept that is very new to the automotive industry. It combines the role of the spring and Anti-Roll Bar into one component. The composite transverse leaf spring should provide very good spring rate all at a reduced weight as compared to a conventional spring system. As long as the spring is correctly designed (internal micro-strain never exceeds a specified level), it promises a maintenance free, high-performance solution for SR8's spring rate. This is very high in innovation but also carries a moderate risk not being available in time.

There is also research being conducted into a regenerative damper system for SR8. This would use the same mounting positions as a conventional damper, but instead of using oil to damp the motion of the wheels, the regen damper would convert the mechanical motion into electricity that could be sent straight back to the battery. On a smooth road this system does not provide much benefit, but over a rougher surface the effects get exponentially more effective. This is an exciting concept because instead of the dampers converting the mechanical movement into waste heat, this energy is recycled in order to increase the cars range/performance. This is very high in innovation but also carries a moderate risk not being available in time.

Component	Innovation	Risk	Overall
Bushings	Low	Low	
Ball-Joints	Low	Low	
Transverse leaf spring	High	Med	
Regen dampers	High	High	
Double wishbone	Med	Low	

2.4 Manufacture and Procurement

Manufacture of the suspension system would be primarily in-house, with certain components where there would be little to no benefit from attempting to produce the component in-house.

Table 4: Manufacture and Procurement

Subsystem	Method

Control Arms	Manufacture in house/Sorensen
Bushings (Rubber)	Source bushings
Bushings (Poly-Urethane)	Manufacture in house, source material
Ball-joints	Source from supplier
Transverse Composite Leaf Spring	Manufacture in house
Dampers	Source from supplier

2.5 Plan Design and Production Plan

Key milestones include:

- Preliminary Design
- Detailed Design
- Construction

2.6 Cost Analysis

A cost analysis is fundamental and will require a more detailed design to calculate a better estimate. A conceptual estimate is provided below

Component	Cost per unit	Approx Qty
Bushings	$20-100	16
Ball-Joints	$20-100	8
Transverse leaf spring	High	2
Regen dampers	High ($1000???)	4
Wishbone	$900	8

2.7 Educational Outcome Analysis

If manufacture were to be viable and safety is a factor that can be upheld, the process of manufacture would provide an exceptional learning opportunity for Vehicle Dynamics and more broadly the Mechanical team.

This learning experience would require communications with McConaughy and potentially Sorensen to ensure that safety items are met.

The students will be able to:

- Design and optimise a suspension geometry
- Research off-the-shelf components to satisfy the functional requirements
- Generate CAD and CAM models as well as engineering drawings for the manufacture of components
- Perform FEA analysis on critical components
- Collaborate efficiently with other departments such as Chassis & Bodywork, and powertrain

2.8 Upskilling Development

While the manufacture of certain components may be outsourced but, inductions in Mill and Lathe would help the team understand the limitations of manufacturing methods, so that they can adjust their designs accordingly. Welding could also prove to be useful depending on the chosen manufacturing methods.

The OptimumG software is a powerful tool that is used heavily in the design of a suspension model. The team must be trained on this software to ensure it is used to its full potential. There are online tutorials available online from OptimumG that clearly explain the software.

- OptimumG Series
    - OptimumDynamics
    - OptimumKinematics

3. Requirements

3.1 Australian Design Rules ADRs

Table 5: ADR requirements

Reference	Title	Description	TPM
REQ4.1	ADR43 Section 6.1.4.1.1	Complies with minimum loaded ride height	TPM 401 – for the mid-point between any 2 consecutive '*Axles*', the dimension in millimetres obtained by multiplying the distance between those 2 '*Axles*' in metres by 33.33
REQ4.2	ADR43 Section 6.1.5.1	Complies with overall vehicle width	TPM 402 – Total vehicle width must not exceed 2,500 mm
REQ4.3	ADR43 Section 6.2.6.2	Complies with minimum unladen ride height	TPM 403 – 'Running' Clearance must not be less than 100 mm.

3.2 Functional Requirements: Normal Operation

Table 6: Functional requirements table (Normal conditions)

Reference	Title	Description	TPM
REQ1.1	Vehicle loading	Sustains full load acting on vehicle.	TPM101 – The full weight of SR8 is sustained by the suspension components.

REQ1.2	Vehicle moving and steering	Allows easy movement and steering of vehicle weight.	TPM102 – Vehicle can move and steer at variable speeds and loading.
REQ1.3	NVH	Isolates the occupant cell from the road to reduce NVH.	TPM103 – The NVH in the cabin is kept as low as possible while still achieving performance requirements
REQ1.4	Vibrations	Structurally withstands all road vibrations.	TPM104 – Wheels handle all road vibrations caused by tarred road travel (Track and public road)
REQ1.5	Braking	Withstands braking forces applied.	TPM105 – Braking forces exerted at variable speeds, temperatures and road condition are withstood in road and track conditions.
REQ1.6	Modularity	Accessible and replaceable within a required timeframe.	TPM106 – Suspension is modular in design and can be replaced within a short time frame with set tooling and torque settings.
REQ1.7	Versatility	Suspension is safe in all weather conditions.	TPM107 - Suspension can maintain functional requirements regardless of weather conditions.
REQ1.8	Maintenance	Suspension components have maintenance schedules and checking procedures for routine checks, pre track, post track and before all long road drives.	TPM108 – Suspension components have clear maintenance schedules and routines that align with use of SR8.
REQ1.9	Self-centering	Ensure that the wheels return to center if no load is applied to the steering wheel	TPM109 – Suspension geometry must exhibit a self-centering force on the wheels.

3.3 Performance Requirements: Teamwide Specifications

Table 7: Performance requirements (Teamwide specifications)

Reference	**Title**	**Description**	**TPM**
REQ2.1	Contact patch	Maintains adequate contact patch between tire and road in all driving conditions.	TPM201 – Tire camber angle must not exceed $\pm 4.5°$ of camber angle.
REQ2.2	Suspension mass	The total mass of the suspension should be kept at a minimum. The suspension components are counted as unsprung mass.	TPM202 – The total mass should be kept as low as possible while satisfying all load cases.
REQ2.3	System COG	Centre of gravity is critical to the handling of SR8.	TPM203 – The suspension components must contribute to the target COG height

REQ2.4	Mass distribution	The closer the suspension mass is located to the center, the easier the car will rotate.	TPM204 – The suspension components are kept as close to the center of the car as possible.
REQ2.5	Track performance	Suspension should optimize performance around the track. In accordance with the SR8 vision.	TPM205 – SR8 must be able to achieve the lap times as laid out in the SR8 vision document.
REQ2.6	Innovation	Suspension design should be innovative to generate interest in the project	TPM206 – Suspension design must include at least one innovative/high performance design.
REQ2.7	Sustainability	Suspension must be as sustainable as possible in terms of material use and manufacturing while still satisfying performance requirements.	TPM207 – Suspension is designed to be sustainable.

3.4 Interface Requirements: Project/System Interface

Table 8: Interface requirements (Project/System interfaces)

Reference	Title	Description	TPM
REQ3.1	Packaging	Interface between projects. Close collaboration with Packaging to enable other departments to interface with suspension and to determine the space in which suspension operate.	TPM301 - Packaging requirements for suspension travel and turning radius are met and allow full range of steering. Including required steering angle.
REQ3.2	Test Mule	Suspension to supply working product to test mule to enable testing of key data.	TPM302 - Test mule has accurate dimensions (Wheelbase and track) for effective testing and optimization. Suspension is supplied to test mule.
REQ3.3	Brakes	Interface is the connection between the suspension and the uprights of the brake system.	TPM303 – Suspension geometry is dependent on the geometry of the uprights, which are within the brakes project.
REQ3.4	Steering	Interface is the connection between the steering inputs and steering behaviour during suspension travel.	TPM304 - Wheelbase, track and key data is supplied for design of steering subsystem.
REQ3.5	Wheels	Interface is in the form of weight transfer and cornering g-force required to design and optimize the wheels subsystem.	TPM305 -Weight transfer, wheelbase, track, and cornering g-force is supplied. Allowing the design and optimisation of the wheels subsystem.
REQ3.6	Exterior	Exterior body panels must not interfere with the suspension at any point in the travel range.	TPM306 – Pass on full suspension travel information to the Exterior design project

REQ3.7	Occupant Cell	Occupant cell must not interfere with the suspension at any point in the travel range.	TPM307 – Pass on full suspension travel information to the occupant cell project
REQ3.8	Subframes	The suspension components mount to the chassis through the subframes	TPM308 – Inform the subframes project of the required mounting points for an ideal suspension geometry
REQ3.9	Active Aero	The suspension will need to be designed to accommodate the extra downforce exerted by the aero components when deployed	TPM309 – Obtain aero information from active aero project and design spring and damper system accordingly.

3.5 External Requirements